3rd run

- Process 2 Ton in ~~24~~ 20 hrs.

$Q_{T1} = 2000 \times 1.98 \times 11$
$= 43560$

$Q_{T2} = 2000 \times 187$
$= 374000$

$Q_{T3} = 38700$

$\sum Q = 456{,}260 + (0.4 \times 456260)$
$\approx 640K$

$\frac{640000}{20 \times 3600} = 8.88 \, KW$
$\simeq 9 KW$

- Fluid: water at 60°C.
- Vessel design passed: Tank A: D = 1.8m, h = 2m; Tank (M_1, M_2, M_3): D = 2.32 m, h = 2.2m.

- V cap = $1.5 ms^{-1}$

Flow rate $= \pi (0.0279)^2 \times \frac{1.5^2}{4} \times (3600)$
$= 0.001376$
$= 4.952 \, Kg/hr$

from 2nd run: L = 100m, 50m (In estimating 50)

d = 1.6m, h = 1.8m, L = 50

$\frac{V_1}{V_2} = \frac{A_2}{A_2}$

$V_1 A_1 = V_2 A_2$

$V_1 = 1.5 \times \frac{d_2^2}{d_1^2} =$

$$L = \frac{h}{p}\left(\sqrt{\pi d^2 + p^2}\right)$$

or, $50 = \frac{1.8}{p}\left(\sqrt{\pi 1.6^2 + p^2}\right)$

or, $50p = 1.8\left(\sqrt{8.0425 + p^2}\right)$

or, $\frac{50p}{1.8} = \sqrt{8.0425 + p^2}$

or, $771.6049\ p^2 = 8.0425 + p^2$

or, $770.6049\ p^2 = 8.0425$

or, $p^2 = 0.0104$

or, $P = 0.102\ m$

$\therefore P = 102.16\ mm$

$$\text{Area} = \pi d l$$
$$= \pi \times 0.0483 \times 50$$
$$= 7.587 m^2$$

For $v = 1.5 ms^{-1}$

$$Re = \frac{985.7 \times 0.0427 \times 1.5}{0.504 \times 10^{-3}}$$
$$= 125266.047$$

$$f(s) = (0.79 \ln(125266.047) - 1.64)^{-2}$$
$$= 0.0172$$

$$F(n) = 0.0172 \times \left[1 + 3.54\sqrt{\frac{Din}{Dn}}\right]$$
$$= 0.0266$$

$$h(c) = 0.0266 \times \frac{50}{0.0427} \times \frac{1.5^2}{19.62}$$
$$= 3.60 m.$$

Nu/F.

$f/8 = \cancel{0.0268}\ 0.00338$

$Pr = 3.26$

$$Nu = \frac{f}{8} \times 124\,266 \times 2.897 \times 1.199 \times 1.038$$

$$= 1492.467$$

$$h_{in} = \frac{1492.467 \times 0.648}{0.0427} = 22649.148$$

$$R_{wall} = 0.0483 \times \frac{\ln(1.292)}{30} = 0.000196$$

$$R_{fin} = 0.000075$$

$R_{fout} =$

	Solid	liquid
	91.5%	8.5%
	0.00075	0.0004

$$= 0.000720$$

$$Ra = 2.975 \times 10^6$$

Nu/F/II. Pr. of Palm oil

91.5%	8.5%
5.000	3.00

$$= 4600.5$$

$$Nu_o = \left[0.6 + \frac{4.641}{1.002}\right]^2 = 5.232$$

$$h_{out} = \frac{5.232 \times 0.168}{0.04883} = 18.200 \rightarrow 94.616$$

$$\frac{1}{U} = 0.0000442 + 0.000196 + 0.000075 + 0.00072 +$$

0.054950

$= \cancel{0.05598}$ $\therefore U = \cancel{17.863}\ 86.17$

$= 0.0116042\dot{6}$

$m = 1.356$

Now tricky NTV,

$U = \cancel{17.863}\ 86.17$

91.5%　　　　　　　　8.5%

$NTV = \dfrac{17.863 \times 7.587}{1.356 \times 4183} = \cancel{0.02389}\ 0.115267$

91.5% of NTU for Phase change = 0.

0.02186

$\varepsilon_1 = 1 - e^{\cancel{0.02186}\ 0.105469}$

$= \cancel{0.02163}\ 0.100098$

8.5% of NTU for liquid zone

$$\varepsilon_2 = \frac{1 - e^{-NTU(1-C_r)}}{1 - C_r e^{-NTU(1-C_r)}}$$

for 0.09

$\varepsilon_2 = 0.002028$

$Q_{max_1} = (0.0013\cancel{7}6 \times 985.7) \times 4183 \times 19$

$= 1.356 \times 4183 \times \cancel{19}\ 35$

$= \cancel{107796.499}\ (0.915)\ \boxed{\cancel{141803.7}}$

1.98525.18 ......

$Q_{max2} = 1.356 \times 4183 \times 10.$

$= 56721.48\ (0.085)$　$= 19.8\ KW$

Finally Max = 98633.796 kw

4821.326 w

After NTU = 2133.449

= 9.7

$\varepsilon = \varepsilon_1 + \varepsilon_2 = 0.023658$

$\therefore Q_{actual} = \cancel{2550.249}\ \cancel{2.5\,KW}$

Not acceptable (while keeping $1\frac{1}{2}''$ pipe)

Retry with $L = 100m$, $1\frac{1}{2}''$ pipe,

$$NTU = \frac{17.863 \times 15.124}{1.356 \times 4183} = 0.04779$$

for Phase change, 91.5%

$NTU = 0.043724$

$\varepsilon = 0.042783$

$Q_{actual} = 6066.787\ KW$

$$NTU = \frac{135.504}{4183n} = \frac{0.03239}{n}$$

$$\varepsilon = 1 - e^{\frac{0.03239}{n}}$$

mass flow rate $\rightarrow Q = \left(1 - e^{\frac{0.03239}{n}}\right) \cdot (n \times 4183 \times \cancel{20}35)$

Area $\rightarrow \cancel{Q = (1 - e^{\cancel{0.6}}}$

$Q = \left(1 - e^{n \cdot 0.003149}\right)(1.41.2037)$

$n = 1, 10, 15, 20$

by using the function, For $20m^2$ we can melt 2014.5 Kg palm oil, in 24 hrs.

$\pi d l = 20$ or, $\pi \times 0.0483 \times l = 20$

$l = 135$

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Accounting for length

$Q = (1 - e^{-NTU}) \times 198.52518$

$NTU = n \cdot 0.008149$

$= \pi d l \times 0.008149$

$= l \times 49.743 \times 10^{-5}$

$\therefore Q = (1 - e^{-l \times 49.743 \times 10^{-5}}) \times 198.52518$

For,

$1.5 ms^{-1}$ fluid velocity passing through

$1\frac{1}{2}''$ pipe, considering palm oil at $25^{\circ}C$

checking if ~~135~~ ~~120~~ 135 m pipe fits in,

$L = 135, \; D_{coil} = 1.6, \; h = 1.9$

$L = \frac{h}{p}(\sqrt{\pi d^2 + p^2})$

or, $135 = \frac{1.9}{p}(\sqrt{8.0425 + p^2}$

or, $5048.476 p^2 = 8.0425 + p^2$

or, $p = 0.0399$ m

$\simeq 40$ mm

~~$L_{max} = 1$~~

$D = 1.7 \quad h = 2, \; L = 135$

$135 = \frac{2}{p}(\sqrt{8.0425 + p^2})$

~~67.5p~~ $4556.25 p^2 = 8.0425 + p^2$

1.765 mm

$$L_{max} = \frac{2}{0.09652} \times \sqrt{8.044829}$$
$$= 117\,m.$$

If stirred, $Re_i = \frac{\rho N D_{ine}^2}{\mu}$

$$Nu = C \cdot Re_i^a \cdot Pr^{1/3} \left(\frac{\mu}{\mu_w}\right)^{0.14}$$

$C \approx 0.4$ to 0.6.
$a \approx 0.5$ to 0.67.

$$h_{in} = \frac{Nu \times K}{D_{pine}}$$

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